

13. Ectopic Pregnancy and Pregnancy of Unknown Location

Companion reading handout for simultaneous listening.

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In early pregnancy, the dangerous mistake can be delay. But the dangerous mistake can also be false certainty.

That is the tension of this lecture. Ectopic pregnancy can rupture, hemorrhage, and kill. But treating too early, before the pregnancy location is reasonably understood, can interrupt or injure a desired intrauterine pregnancy.

So the dominant model is simple, and it should sit at the front of your mind every time you see a positive pregnancy test with pain, bleeding, or an empty uterus on ultrasound.

Location before medication.

And the paired rule is just as important. Stable uncertainty is allowed only when follow-up is tight.

Start with why ectopic pregnancy deserves respect. Ectopic pregnancy means implantation outside the uterine cavity. More than ninety percent are tubal. It accounts for about two percent of reported pregnancies, but that number understates the risk in the patients who actually present to emergency care. Among patients with first-trimester bleeding, abdominal pain, or both, ectopic prevalence has been reported as high as eighteen percent.

Rupture remains a major cause of morbidity and mortality. The tube is not built to host a growing trophoblast. As the pregnancy invades and expands, the clinical danger is not just the hCG number. The danger is bleeding from a structure that cannot safely stretch and cannot safely contain placental invasion.

Now risk factors help you raise suspicion, but they do not let you relax when they are absent.

Classic risk factors include prior ectopic pregnancy, tubal damage, ascending pelvic infection, prior pelvic or tubal surgery, assisted reproductive technology, infertility itself, cigarette smoking, and age older than thirty five.

A history of ectopic pregnancy matters especially. After one prior ectopic, recurrence risk is about ten percent. After two or more, it rises to more than twenty five percent.

But half of patients diagnosed with ectopic pregnancy have no known risk factor. That is the practical reason every reproductive-aged, sexually active patient with

abdominal pain or vaginal bleeding deserves pregnancy testing, even if she is using contraception.

The intrauterine device is the classic trap. An IUD lowers the absolute risk of ectopic pregnancy because it prevents pregnancy very effectively. But if pregnancy occurs with an IUD in place, the proportion that are ectopic is high, up to fifty three percent in the ACOG source. So the sentence is not, IUD means ectopic is likely in all users. The sentence is, pregnancy with an IUD in place must be located carefully.

The first clinical fork is not hCG. It is stability.

If the patient is hemodynamically unstable, has an acute abdomen, or has signs concerning for rupture, urgent evaluation and treatment take priority. Do not let a diagnostic algorithm slow down resuscitation and surgery when the patient is declaring rupture.

If she is stable, you earn the right to diagnose carefully.

The minimum evaluation for suspected ectopic pregnancy is confirmation of pregnancy and transvaginal ultrasound. Often that first evaluation is not enough. Serial transvaginal ultrasound, serial hCG, or both may be needed.

Now listen carefully to the language of ultrasound, because the wording matters.

A definitive ectopic pregnancy on ultrasound is an adnexal gestational sac with a yolk sac or embryo. That is definitive because you have pregnancy structures outside the uterus.

But most ectopic pregnancies do not arrive with that perfect image. More often, ultrasound shows a mass, or a mass with a hypoechoic area, separate from the ovary. That should raise suspicion, but it is not the same as seeing an extrauterine sac with a yolk sac or embryo. The positive predictive value is about eighty percent, because other pelvic structures can mimic ectopic pregnancy: corpus luteum, hydrosalpinx, endometrioma, paratubal cyst, or bowel.

The resident reflex is to respect suspicious ultrasound findings without pretending they are always definitive.

Intrauterine Findings

A definitive intrauterine pregnancy is a gestational sac with a yolk sac or embryo. Seeing that usually excludes ectopic pregnancy.

The exception is heterotopic pregnancy: an intrauterine pregnancy and an ectopic pregnancy at the same time. It is rare in naturally conceived pregnancies, roughly one in four thousand to one in thirty thousand, but after in vitro fertilization the risk can be as high as one in one hundred. That is why assisted reproduction changes your threshold for caution.

Do not overcall a sac-like fluid collection. A hypoechoic sac-like structure in the uterus, even something that looks tempting, can be a pseudogestational sac. It may be fluid or blood in the uterine cavity in a patient who actually has an ectopic pregnancy. The words yolk sac or embryo matter because they are the structures that turn suspicion into location.

hCG

hCG is useful. hCG is not a map.

A single hCG value cannot diagnose viability and cannot diagnose location. It must be interpreted with the patient's symptoms, history, and ultrasound findings.

The most famous hCG concept is the discriminatory level. This is the hCG value above which landmarks of a normal intrauterine pregnancy should usually be visible on transvaginal ultrasound.

But the discriminatory level is one of the easiest numbers to misuse.

If no intrauterine gestation is seen above the discriminatory level, the pregnancy is very concerning for a nonviable gestation, either early pregnancy loss or ectopic pregnancy. In many cases, those findings are consistent with ectopic pregnancy. But the discriminatory level has been challenged because viable intrauterine pregnancies have been documented on follow-up after no sac was seen on the initial scan even though the hCG was above older discriminatory thresholds.

So if you use the discriminatory level, use it conservatively. ACOG says the value should be conservatively high, for example as high as three thousand five hundred milli international units per milliliter, to avoid misdiagnosis and possible interruption of an intrauterine pregnancy that the patient hopes to continue.

The mechanism is straightforward. hCG is a biologic signal, not a clock with perfect timing. Dating can be wrong. Normal variation exists. Multiple gestations can have higher hCG values before the expected ultrasound landmarks are visible. If you choose a low threshold and treat absence of a sac as proof, you can harm the exact pregnancy you were supposed to protect.

Serial hCG helps, but it also has limits.

When abnormal gestation is suspected, obtain a second hCG two days after the first. Subsequent measurements are usually every two to seven days, depending on the pattern and the level of change.

In a potentially viable intrauterine pregnancy, the minimum expected rise over forty eight hours depends on the starting hCG. If the initial value is less than fifteen hundred, the minimum expected rise is about forty nine percent. If it is fifteen hundred to three thousand, it is about forty percent. If it is greater than three thousand, it is about thirty three percent.

These cutoffs teach a deeper point. The higher the starting hCG, the slower the expected rise. So do not memorize one old doubling rule and use it everywhere. Early pregnancy biology is curvilinear.

A rise below those minimum thresholds is suspicious for abnormal pregnancy, either ectopic pregnancy or early pregnancy loss. But it does not prove location.

A falling hCG suggests a failing pregnancy. But it still does not prove location. A patient with decreasing hCG values and possible ectopic pregnancy must be followed until nonpregnant levels, because rupture can occur while hCG values are falling or very low.

That sentence is clinically important. Falling hCG lowers anxiety, but it does not erase rupture risk.

We can define pregnancy of unknown location

Pregnancy of unknown location means the pregnancy test is positive, and ultrasound shows neither a definitive intrauterine pregnancy nor a definitive ectopic pregnancy.

It is not a diagnosis. It is a temporary state.

This is more than semantics. If you call pregnancy of unknown location a diagnosis, you may feel licensed to treat it as ectopic pregnancy. But it may be an early intrauterine pregnancy, a failing intrauterine pregnancy, or an ectopic pregnancy that is not yet visible. The name means location has not yet been established.

For a clinically stable patient who would continue the pregnancy if it is intrauterine, the usual safe path is repeat transvaginal ultrasound, serial hCG, or both. Follow-up is not passivity. It is the treatment of uncertainty with evidence.

This is where residents sometimes feel uncomfortable. The patient is anxious. The clinician is anxious. The ultrasound is nondiagnostic. The hCG is not reassuring. Everyone wants a label.

But anxiety is not an indication for methotrexate.

Methotrexate is treatment. It is not a diagnostic test.

Unnecessary methotrexate can interrupt an ongoing intrauterine pregnancy or create teratogenic exposure. That is why ACOG emphasizes follow-up to confirm the diagnosis in stable patients, especially at the first clinical encounter, when misdiagnosis risk is highest.

There is one diagnostic tool that can help when the question is failed intrauterine pregnancy versus ectopic pregnancy: uterine aspiration.

But timing matters. Uterine aspiration is considered when a progressing intrauterine pregnancy has been reasonably excluded, and when the decision fits the patient's

values about maternal risk and the possibility of interrupting a pregnancy. It is not something to use casually when the patient still hopes for a viable intrauterine pregnancy and viability has not been reasonably excluded.

The logic is histologic.

If chorionic villi are found after aspiration, you have confirmed intrauterine trophoblast. In that setting, the diagnosis is failed intrauterine pregnancy, and ectopic pregnancy is effectively excluded except for the rare heterotopic scenario.

If chorionic villi are not found, you have not automatically diagnosed ectopic pregnancy. You have created a new fork.

The next step is hCG measurement twelve to twenty four hours after aspiration.

If the hCG plateaus or rises, that suggests incomplete evacuation or a nonvisualized ectopic pregnancy, and further treatment is warranted. A practical definition of plateau is a decrease of less than about ten to fifteen percent, although the exact boundary is not perfectly defined.

If hCG decreases by fifty percent or more within twelve to twenty four hours, that strongly suggests failed intrauterine pregnancy. But even then, follow-up remains important, because rare ectopic pregnancies can show substantial early declines.

If the decline is between fifteen and fifty percent, do not pretend the result is clean. That middle zone requires close serial follow-up and individualized management, using symptoms, risk factors, clinical suspicion, and whether the hCG later plateaus or rises.

This is the heart of the lecture. Ectopic diagnosis is not built from one dramatic number. It is built from stability, ultrasound structures, hCG behavior, time, and sometimes tissue.

Let's walk through the common traps.

A patient has pain, a positive pregnancy test, and no intrauterine pregnancy on ultrasound. That is not automatically ectopic pregnancy. First ask whether she is stable. Then ask whether there is a definitive ectopic finding. Then ask whether she desires continuation if the pregnancy is intrauterine. Then use serial hCG and repeat ultrasound if stable.

A patient has an hCG of twenty two hundred and an empty uterus. That may be deeply concerning. But if she is stable and the pregnancy is desired, you should not use an older low discriminatory threshold as a weapon against a potentially viable intrauterine pregnancy.

A patient has a sac-like intrauterine fluid collection. Do not call it a definitive intrauterine pregnancy unless you see a yolk sac or embryo.

A patient has falling hCG. Do not discharge her from your mind. Follow to nonpregnant levels if ectopic pregnancy remains possible.

A patient conceived with in vitro fertilization and has an intrauterine pregnancy on ultrasound but still has significant pain or concerning adnexal findings. Remember heterotopic pregnancy. Rare is not never, especially after assisted reproduction.

And the biggest trap: a stable patient has pregnancy of unknown location, and the team wants to give methotrexate so the uncertainty ends. That is backwards. The uncertainty is exactly why methotrexate may be unsafe. Location before medication.

Final Synthesis

Ectopic pregnancy diagnosis is a disciplined safety process. If the patient is unstable, act. If she is stable, do not let urgency turn into false certainty.

Use ultrasound to look for definitive structures. Use hCG as a trend, not a map. Use a conservative discriminatory level if you use one at all. Treat pregnancy of unknown location as a transient state requiring follow-up, not as a synonym for ectopic pregnancy. Use uterine aspiration only when the possibility of a progressing intrauterine pregnancy has been reasonably excluded and the result will truly guide care.

Resident Reflex

First decide unstable or stable. Then decide whether location is known or unknown. If location is unknown and the patient is stable, follow tightly. And never use methotrexate simply because uncertainty feels uncomfortable.